Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm:

FIRST SOLVE

■□□ 1/3 · B

4 YELLOW CROSS · yellow edges only — ignore the corners

Dot no yellow edge at all
FRUR'U'F'

Hook two at a right angle: hold them BACK and LEFT
FRUR'U'F'

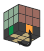
Line hold the bar left-to-right
FRUR'U'F'

One algorithm, up to three times: dot to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn U so at least two top edges match the center under them. This algorithm is called Sune; you use it on every card after this one.

two matched, side by side hold them BACK and LEFT
RUR'U RU2R'


two matched, opposite run it once from any hold, then look again
RUR'U RU2R'

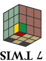
6 PUT THE YELLOW CORNERS HOME

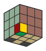
one corner home hold it FRONT-LEFT
RU'L'U R'U'L

no corner home run it once from any hold, then look again
RU'L'U R'U'L

This cycles the other three corners into place. It also twists the yellow face you just built — expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS


yellow faces RIGHT
R'D'RD x2


yellow faces FRONT
D'R'DR x2

Between corners turn ONLY the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look destroyed while you do this. Keep going — the last top turn brings it back.

LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired.
UNLOCK CARD 2 — five scrambles in a row with this card face down. ☐ MASTER: those five under 3:00. My target: ______ · NEXT: 2/3 TWO-LOOK

20 mm

TWO-LOOK

■■□ 2/3 · B

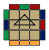
PLL CORNERS · headlights = two matching corners on one face

T headlights on ONE face — hold that face LEFT
RUR'U' R'F R2U'R'U' RUR'F'

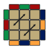
Y no headlights — the two swapping corners sit diagonally
FRU'R'U' RUR'F' RUR'U' R'FRF'

PLL EDGES · corners home, top still not solved

Ub the front edge travels LEFT
R2U RUR'U' R'U' R'UR'


Ua the front edge travels RIGHT
M2U MU2 M'UM2

H both pairs swap straight across
M2U'M2 U2 M2U'M2


Z each pair swaps with its neighbour
M'U' M2U'M2U' M'U2 M2U

START HERE · two new algorithms finish any last layer

Yellow cross: F R U R' U' F' until the cross appears — at most 3.
Yellow face: Sune, read again, repeat — at most 3 (see the badges).
Corners: T-Perm. No headlights to hold left? Run it twice.
Edges: Ub. No solved edge to put at the back? Run it twice.
Each case on the front replaces one repeat with one run. Three a week; you can solve the whole time.

H VS Z

Both show headlights on all four faces. The edge between them decides: it belongs to the opposite face = H, to a neighbour = Z.

STUCK?

Corners home but nothing solves? Turn the top face once and read again — that is often the whole fix.
One case has beaten you five times? Park it, use its fallback above, come back next week.
A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and reseat it.

WORDS

OLL = make the whole top face yellow · PLL = slide the top pieces home · headlights = two matching corners on one face · AUF = the free top turn that lines a case up

THE NEW ORDER: yellow cross, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas: it moves corners but wrecks the yellow face, and here the yellow face is finished first.
UNLOCK CARD 3 — fifteen last layers in a row, each finished in exactly two algorithms, case named out loud first. A repeat does not count. ☐ MASTER: solves averaging under 1:00. My target: ______ · NEXT: 3/3 ONE-LOOK PLL

20 mm

FIRST SOLVE

■□□ 1/3 · A

New here? Do the walkthrough once at cubepath-six.vercel.app/learn. This card is the memory jog.

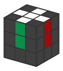
NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube.
' = counter-clockwise. 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit

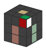
1 WHITE CROSS

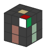


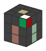
White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm — work it out.
Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.

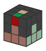

white sticker RIGHT
RUR'U'

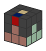

white sticker FRONT
L'U'LU


white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.


goes RIGHT
U RUR'U' y L'U'LU


goes LEFT
U' L'U'LU y' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

🟥 **RUR'U'** sexy move 🟩 **RUR'U** Sune 🟦 **R'FRF'** sledgehammer gap = change grip

---

TWO-LOOK

■■□ 2/3 · A

OLL CROSS · yellow edges

Line bar left-to-right
FRUR'U'F'

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS · yellow face

Sune 1 1 yellow, rest clockwise
RUR'U RU2R'

Anti-Sune 2 1 yellow, rest counter-clockwise
RU2R' U'RU'R'

Pi 2 0 yellow, headlights left
fRUR'U'f' FRUR'U'F'

No yellow edge at all? Run the first one, then read again.

OLL CORNERS · continued

Headlights 3 2 yellow, headlights at the back
R2DR'U2 RD'R'U2R'

2x Headlights 2 0 yellow, headlights both sides
RUR'U RU'R' URU2R'

Chameleon 3 2 yellow next to each other
rUR'U'r' FRF'

Bowtie 3 2 yellow diagonal
F'rUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked; three is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

🟥 **RUR'U'** sexy move 🟩 **RUR'U** Sune 🟦 **R'FRF'** sledgehammer gap = change grip

---

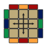
ONE-LOOK PLL

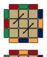
■■■ 3/3 · B

EDGES ONLY · corners already home

● **Ua** B solid; F L R headlights; 3 edges counter-clockwise
M2U MU2 M'UM2

● **Ub** B solid; F L R headlights; 3 edges clockwise
R2U RUR'U' R'U' R'UR'


● **H** all 4 headlights; opposite edges swap
M2U'M2 U2 M2U'M2


● **Z** all 4 headlights; adjacent edges swap
M'U' M2U'M2U' M'U2 M2U

HOW TO LOOK

1. Count the headlight faces. One = front of this card. Four = this column. None = the column on the right.
2. Only then read the edges to pick the exact case.
3. Line the case up as drawn, run it, turn the top again to finish — that last free turn is the AUF.

● = already yours from Card 2.

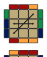
WORDS

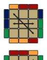
PLL = slide the top pieces home · headlights = two matching corners on one face · AUF = the free top turn that lines a case up · adjacent corner swap = the swapping corners share an edge · diagonal corner swap = the swapping corners sit opposite

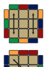
DIAGONAL CORNER SWAP · no headlights anywhere

● **Y** pairs F-left R-back
FRU'R'U' RUR'F' RUR'U' R'FRF'

V pairs F-left L-front
R'UR'U' y R'F' R2U'R'U R'FRF


Na pairs B-left F-right L-front R-back
RUR'U RUR'F' RUR'U' R'F R2U'R' U2RU'R'


Nb pairs B-right F-left L-back R-front
R' URU'R' F'U'F RUR' FR'F' RU'R


E nothing matches
x' L'ULD' L'U'LD L'U'LD' L'ULD

STUCK?

Case you have not learned? Two-look it from Card 2 — T-Perm then Ub still solves it. Nothing here can strand you.
Beaten five times by one case? Park it, two-look it, come back next week.

Card 2 gives you 19/72 last layers in one look. This card gives 71/72.
Next: the other 57 top-face cases, and the first two layers solved as pairs — drilled with a cube in the app: cubepath-six.vercel.app/practice
DONE — all twenty-one named correctly, twenty-one in a row, no card in hand, on two separate days. ☐ MASTER: PLL under 3 s, solves under 0:30. My target: ______

20 mm

ANNEX — BIG CUBES AND THE MAP

□□□ A · B

THE LADDER

1/3 FIRST SOLVE · 9 algorithms · unlock: five solves, card face down.
2/3 TWO-LOOK · +13 · unlock: fifteen last layers in exactly two algorithms.
3/3 ONE-LOOK PLL · +15 · done: all 21 named, twice on separate days.
After that: full OLL, F2L, cross planning, look-ahead — in the app, not on card stock.

WHY THERE ARE ONLY THREE

A card is good at a closed set of cases you tell apart by sight. It is bad at a skill you drill. F2L, cross planning and look-ahead are drills with no finite case list. Full OLL is 57 cases, 22 of which differ only in a sliver a fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised setup. This set stops where the medium stops.

WORDS

OLL = make the whole top face yellow · PLL = slide the top pieces home · Sune = the one-yellow-corner algorithm, used on every card after Card 1 · sexy move = the R U R' U' trigger · sledgehammer = the R' F R F' trigger · headlights = two matching corners on one face · trigger = a short algorithm your hands run as one unit · AUF = the free top turn that lines a case up
parity = a case a 3x3 cannot have · edge pair = two edges glued into one, on a 4x4 or 5x5 · F2L = first two layers, solved as pairs · look-ahead = watching the next piece while your hands finish this one · adjacent corner swap = the swapping corners share an edge · diagonal corner swap = the swapping corners sit opposite

PRINT AND VERSION

Every ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail rather than ship. Recognition wording is generated from the same permutation the diagram is drawn from.
Set v1.0 · reprint any single card at cubepath-six.vercel.app/print · print at 100% / Actual size, never 'Fit to page'.

Not a tier, no unlock, no order. Use it when you buy a 4x4.

20 mm

ONE-LOOK PLL

■■■ 3/3 · A

ADJACENT CORNER SWAP · exactly one face shows headlights

● **T** L headlights; pairs B-left F-left
RUR'U' R'F R2U'R'U' RUR'F'

Ja L solid; pairs B-right F-left R-front
x R2FRF' RU2 r'Ur U2

Jb L solid; pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2U'R'

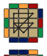
Aa L headlights; pairs F-right R-front
x L2D2 L'U'L D2 L'UL'

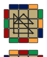
Ab L headlights; pairs B-right R-back
x' L2D2 LUL' D2 LU'L

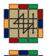
F L solid
R'U'F' RUR'U' R'F R2U'R'U' RUR'UR

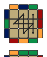
One face shows headlights: two corners of that face match. Hold as drawn, run, then turn the top to finish.

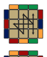
ADJACENT CORNER SWAP · continued

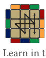

Ra L headlights; pairs F-left
RU'R'U' RUR DR'U'RD' R'U2R'


Rb L headlights; pairs B-left
R2F RURU'R' F'R U2R'U2R


Ga L headlights; pairs F-right
R2UR'UR'U' RU'R2 U'D R'URD'


Gb L headlights; pairs R-back
R'U'R UD' R2UR'URU' RU'R2D


Gc L headlights; pairs B-right
R2U'RU'RU R'UR2 UD' RU'R'D


Gd L headlights; pairs R-front
RUR' U'D R2U'RU'R' UR'UR2D'

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day — one is the mirror of the other.

🟥 **RUR'U'** sexy move 🟩 **RUR'U** Sune 🟦 **R'FRF'** sledgehammer gap = change grip

---

ANNEX — BIG CUBES AND THE MAP

□□□ A · A

BIG CUBES · 4x4 and 5x5

Reduce: build the 6 centers ⟶ pair the 12 edges ⟶ solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY — never widen it.

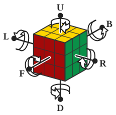
last 2 edges R U R' F R' F' R slice out, run it, slice back – both cubes
4x4 PLL parity 2R2 U2 2R2 Uw2 2R2 Uw2 2 edge pairs swapped · half the time

4x4 OLL parity – 1 edge pair flipped · half the time
Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'

5x5 edge parity – 1 edge pair flipped · half the time · one move differs; 5x5 has no PLL parity
Rw U2 x Rw U2 Rw U2 3Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'

Fix parity during the last layer, **before** OLL and PLL – both parity algorithms move corners.

NOTATION



Each letter turns that face clockwise, seen from outside the cube — so from your seat L, D and B look counter-clockwise. ' = counter-clockwise. 2 = half turn.
M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase r f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT · use it until you know Sune

Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT R'D'RD x2
yellow faces FRONT D'R'DR x2

WORDS

algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit · sexy move = the R U R' U' trigger · sledgehammer = the R' F R F' trigger · parity = a case a 3x3 cannot have · edge pair = two edges glued into one, on a 4x4 or 5x5

🟥 **RUR'U'** sexy move 🟩 **RUR'U** Sune 🟦 **R'FRF'** sledgehammer gap = change grip